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BIOTECHNOLOGY

Cytotoxicity of Laser-Synthesized Nanoparticles of Elemental Bismuth

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High X-ray absorption combined with photothermal properties make bismuth nanoparticles (Bi NP) a promising agent for multimodal cancer theranostics. However, the synthesis of Bi NP by the "classical" chemical methods has numerous limitations, including potential toxicity of the produced nanomaterials. Here we studied *in vitro* toxicity of laser-synthesized Bi NP coated with Pluronic F-127 on mouse fibroblast cell line L929. The survival of L929 cells decreased linearly with increasing the concentration of Bi NP in a concentration range of 3-500 μg/ml; the LC_{50} value was 57 μg/ml. The unique combination of functional properties and moderate toxicity of the laser-synthesized Bi NP makes them a new promising platform for sensitization of multimodal cancer theranostics.

Key Words: *cytotoxicity; laser ablation; bismuth; nanoparticles; fibroblasts*

In recent years, nanotechnology has been actively integrated into biomedicine, especially in the field of oncology. The interest in biomedical nanotechnologies is due to unique properties of nanomaterials, which open new areas of their application both in diagnostics (contrast and sensitization of radiation diagnostics and photoacoustic imaging) [1-5] and in therapy (radiation therapy, photohyperthermia) [5-10] of oncological diseases. Bismuth (Bi) nanoparticles (NP) are of particular interest in this fields. The highest atomic number of Bi among non-radioactive materials (Z=83) and the extremely high X-ray attenuation coefficient (7.38 cm^2/g at 90 keV) define the prospects of Bi NP application for sensitization in radiation theranostics (therapy+diagnostics) [11-13]. The metallic properties of Bi open up opportunities for its applications in photothermal therapy [5,14] and photoacoustic imaging [5-10], which in combination with radiation exposure, can significantly enhance the theranostic effect of NP [10].

Traditional methods for the Bi NP fabrication are based on complex multistep chemical synthesis technologies often requiring the use of toxic precursors. Pulsed laser ablation in liquids (PLAL) is an alternative method for Bi NP synthesis free from the limitations of chemical synthesis. This method allows obtaining ultrapure spherical NP with controllable properties from almost any inorganic material [14-17]. The flexibility, simplicity, and high efficiency of PLAL make the method popular for the synthesis of nanomaterials that can be applied in biomedicine [15,17]. However, the clinical applications of nanomaterials require surface modification of NP with biocompatible polymers such as polyethylene glycol or Pluronic [6,15]. Despite

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the developed methods of synthesis and surface modification, the biocompatibility of laser-synthesized Bi NP still remains unclear.

Here we explore the cytotoxicity of laser-synthesized Bi NP with surface modification with Pluronic F-127 in a series of *in vitro* experiments on mouse fibroblast cell line L929.

MATERIALS AND METHODS

Synthesis of NP. Bi NP were synthesized by femtosecond PLAL in acetone. A metallic Bi target (99.999% purity; Goodfellow) was mounted vertically in a glass cuvette filled with 15 ml of acetone. An Yb:YAG femtosecond laser TETA-10 (λ=1030 nm, pulse duration 270 fsec, pulse frequency 200 kHz, pulse energy 15 μJ, Avesta) was used for ablation of the Bi target for 20 min. Laser radiation was focused on the target surface by a Φ-theta lens with a 100 mm focal length. The thickness of the liquid layer between the inner surface of the cuvette and the target surface was 2.5 mm. The laser beam was continuously moved along the target surface at a speed of 4 m/sec by a galvanometric scanner to increase the productivity of synthesis and to minimize laser energy scattering on cavitation bubbles formed in the process of ablation. The scanning geometry was a double closed spiral with a diameter of 10 mm. The surface modification of the Bi NP was performed with Pluronic F-127 according to the standard protocol described in our previous study [15].

Characterization of the Bi NP. The morphology and size of the Bi NP were determined using a MAIA 3 scanning electron microscope (Tescan). The obtained images were analyzed in ImageJ software. A Zetasizer Nano ZS analyzer (Malvern Instruments) was used to determine the hydrodynamic size distribution of Bi NP by dynamic light scattering, and the ζ-potential was studied by electrophoretic light scattering with Smoluchowski approximation. To obtain the size distribution, the scanning electron microscopy (SEM) images were analyzed using the ImageJ software in round approximation of NP.

Cell line. The mouse fibroblast cell line L929 (ATCC) was chosen based on the potential clinical application of Bi NP for cancer theranostics. The cells were cultured under standard conditions (37°C and 5% CO_2) in DMEM medium containing 10% fetal bovine serum, 100 μM penicillin/streptomycin, and 2 mM L-glutamine.

For analysis of Bi NP cytotoxicity, L929 cells (5×10^3/well) were seeded into a 96-well plate and cultured in supplemented DMEM medium for 24 h. Then, various concentrations of Bi NP were added to the wells for another 48 h. Cells without Bi NP were used as a control. Then, 20 μl of MTT solution (5 mg/ml) was added to each well and incubated for 2 h at 37°C and 5% CO_2. The MTT medium was aspirated, after which 100 μl of DMSO was added to the wells to dissolve the formed formazan salts. The absorbance of the resulting solution was measured at wavelengths of 570 and 630 nm on a M1000 microplate spectrophotometer (Tecan). The obtained values were normalized to the control. All experiments were performed in at least 6 repetitions.

Statistics. Statistical processing and calculation of the mean values were performed using the OriginPro 18 program (OriginLab). The significance of the differences was assessed using Student's *t* test and was considered statistically significant at $p<0.05$. The data are presented as $M\pm SD$.

RESULTS

Laser ablation of the crystalline Bi target yielded a stable colloidal solution of spherical Bi NP (Fig. 1, *a*). The optimal size of NP for biomedical applications is ~50 nm [17], which ensures effective internalization of NP by cells and increases the duration of their bloodstream circulation. Dynamic light scattering analysis of the hydrodynamic diameter distribution of Bi NP showed that the mode of the size distribution was 50 nm with a full width at half maximum of 28 nm (Fig. 1, *c*). These results are in good agreement with the data of SEM image analysis (Fig. 1, *b*). At the same time, this size of the laser-synthesized Bi NP corresponded to the optimal range for biomedical applications that was mentioned above (Fig. 1, *b*). The size distribution had a lognormal character.

ζ-Potential of Bi NP with biocompatible polymer Pluronic F-127 surface-modification was 19.7±8.0 mV (Fig. 1, *d*). The high value of the double layer potential of the modified Bi NP determines their high colloidal stability in aqueous medium, which is a key factor for their further application in biomedical purposes. It is also worth noting that the presence of a positive charge ζ-potential allows one to give the NP new functionalities by their conjugation with bioactive molecules (*e.g.*, targeting antibodies specific to antigens of a particular cancer cell line), which increases the selectivity of binding of the theranostic agent to the target sites.

The *in vitro* study of the cytotoxicity of surface-modified Bi NP revealed the dependence of L929 cell line survival on the concentration of Bi NP (Fig. 2). LC_{50} value was 57 μg/ml, which exceeds the similar values for chemically synthesized Bi NP (50 μg/ml) [18]. At the same time, the dependence of survival rate on the concentration of Bi NP was linear. It should be noted that the probable mechanism of cell death

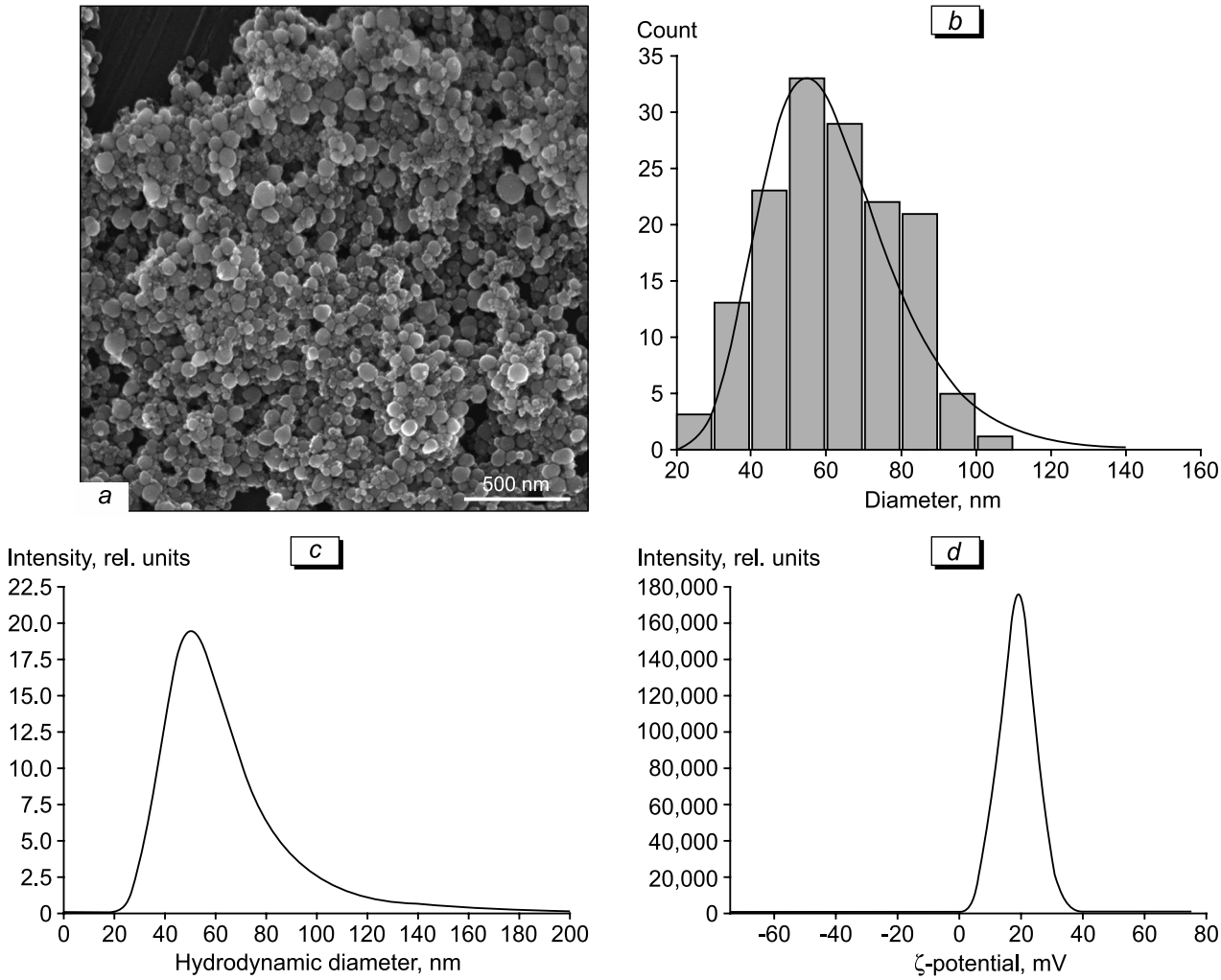

Fig. 1. Bi NP with surface modification with Pluronic F-127, SEM (*a*); size distribution (*b*), distributions of the hydrodynamic diameter (*c*), and ζ-potential (*d*) of laser-synthesized Bi NP.

initiation in this case could be oxidative stress due to enhanced oxidation of the Bi NP surface under biologically relevant conditions and the following local ROS generation [19,20]. Moderate cytotoxicity of laser-synthesized Bi NP with surface-modification with Pluronic F-127 in combination with their X-ray absorbance and photothermal properties open up new promising opportunities for the development of multimodal platform for advanced theranostics of oncological diseases.

Thus, a stable colloidal solution of Bi NP was obtained by PLAL. The obtained NP had a spherical morphology, narrow size distribution with a mode in the range of 50-60 nm with a full width at half maximum of 33 nm, as well as a positive ζ-potential value of 19.7±8 mV. These parameters of NP are optimal for their biomedical applications. We also studied the *in vitro* toxicity of Bi NP with surface-modification with Pluronic F-127 Bi NP on L929 cell line and determined LC_{50} value (57 μg/ml). Low cytotoxicity and high theranostics potential of laser-synthesized Bi NP determines the prospects for their application for solving the most pressing problems of modern

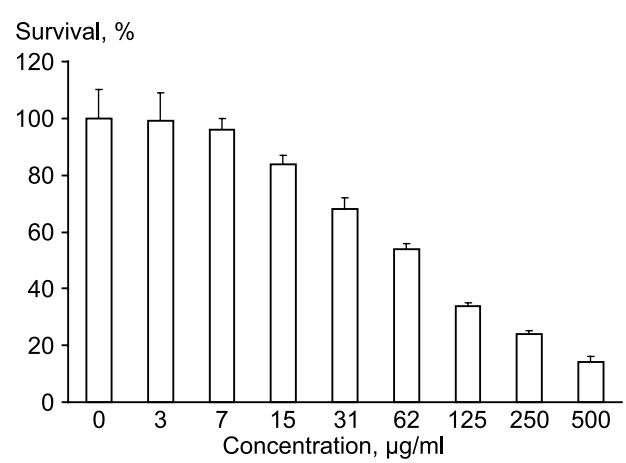

Fig. 2. Survival of mouse fibroblast line L929 cells in the presence of different concentration of laser-synthesized Bi NP.

biomedicine. At the same time, moderate toxicity in combination with multifunctionality of Bi NP forms a reliable platform for effective sensitization of cancer theranostics.

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Conflict of interest. The authors have no conflicts of interest to declare.

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